

Gerai 1000 Pintu — AI Department

Arsitektur, Memori, Workflow, dan Roadmap Pengembangan

Versi	1.0
Tanggal	24 Mei 2026
Pemilik	Matthew (mattlouai@gmail.com)
Status	PRODUCTION LIVE di gerai.maheetwork.com
Mode	Solo-tenant — hanya Matthew yang pakai

Daftar Isi

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1. Ringkasan Eksekutif

Gerai 1000 Pintu adalah platform retail destination "Dunia Pintu Indonesia" yang dibangun Matthew sebagai solo founder. Untuk mengoperasikan bisnis multi-pilar (Product, Knowledge, Service) dengan tim kecil (lean store 2 orang per toko + Door Expert generalist terpusat), Matthew bangun **AI Department** sebagai partner berpikir yang menggantikan fungsi tim level menengah-atas.

AI Department = Atmaja (CEO) + 4 C-suite (orchestrator) + 12 specialist, berjalan di atas stack web app modern (Vite/React PWA + Vercel serverless + OpenRouter LLM).

Status saat ini (24 Mei 2026):

- LIVE** PWA, Atmaja chat dengan vision + native PDF reading
- LIVE** Memory persistent 3-lapis (working memory 100 pesan, long-term file KV, PDF library Blob)

- LIVE

 2 n8n workflow (Brief Lifecycle synchronous + Daily Digest scheduled)
- LIVE

 12 endpoint server-hardened (strict origin + bearer auth)
- PENDING

 2FA, channel output Daily Digest, workflow #3-5

Cost steady-state estimate: \$30-45 per bulan untuk solo intensive use.

2. Identitas & Konteks Bisnis

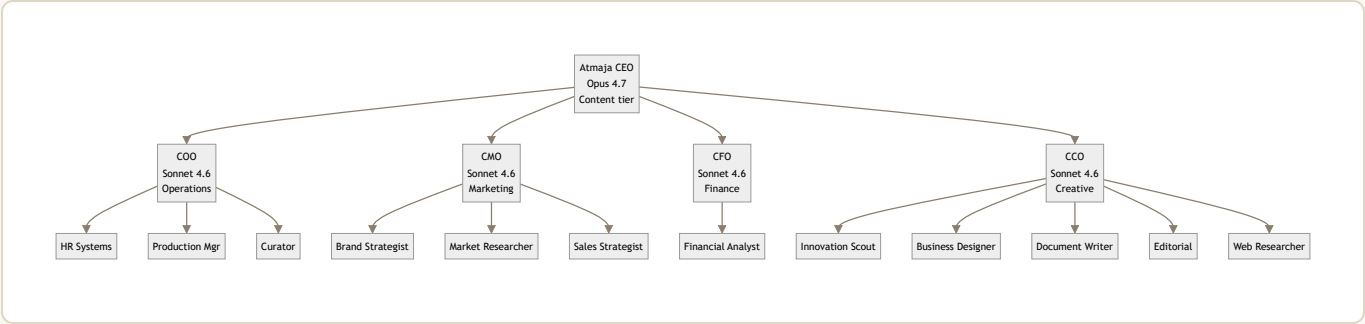
2.1 Bisnis Gerai 1000 Pintu

Aspek	Detail
Nama lengkap	Gerai 1000 Pintu
Tagline	Dunia Pintu Indonesia
Entitas hukum	PT Selaras Lawang Sewu (SLS)
Brand pertama	AMK (di bawah SLS)
Lokasi pilot	Balikpapan, Kalimantan Timur
Founder	Matthew
Tim	Solo (Matthew) + AI Department
3 Pilar	Product (katalog) · Knowledge (edukasi) · Service (Door Expert + Cash & Delivery)
Differentiator	Lean Store 2 orang, Self-Ordering Kiosk, Konsultasi Terpusat
Brand canon	Calm refined premium curated retail. NO em-dash. "Tempat" bukan "rumah". Nama "Gerai 1000 Pintu" lengkap.
Target Q3 2026	Launching AMK wave 1 di Balikpapan minggu ke-2 Juli
Vendor pintu kayu	PT Selaras Lawang Sewu (lead time 21 hari, NET 30)
Budget marketing wave 1	Rp 50 juta (Meta Ads + IG influencer mikro Kaltim)

2.2 Aplikasi Gerai

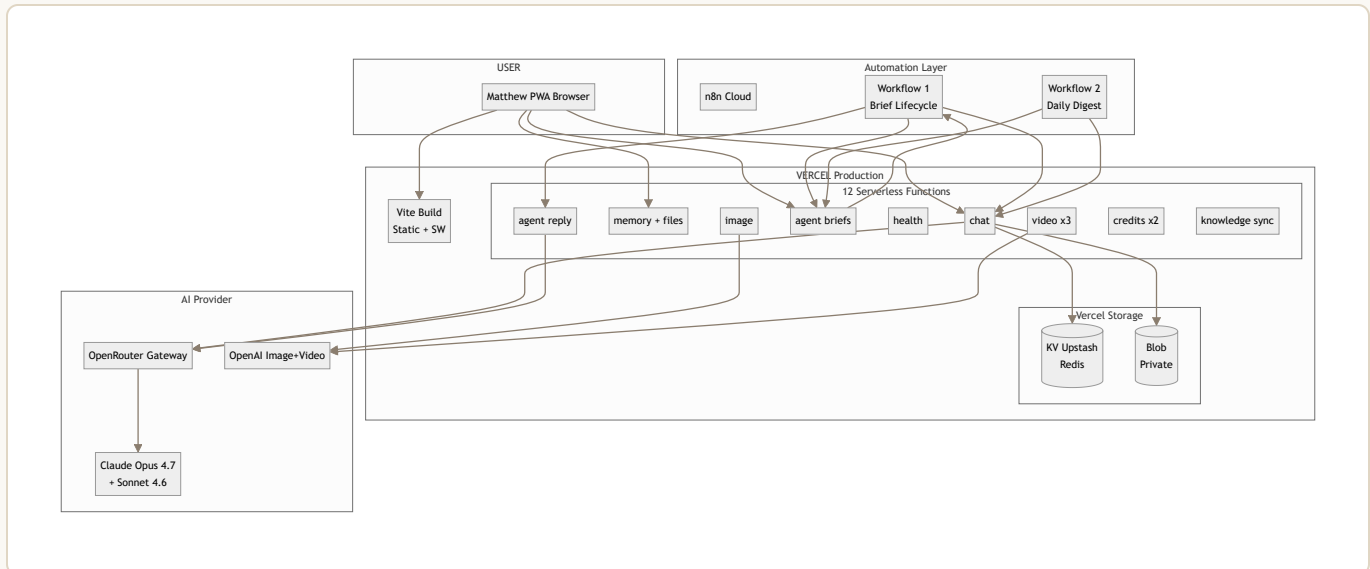
Aspek	Detail
Production URL	gerai.mahework.com
Repo	github.com/MahewAi/Mahew (private, branch main)
Tech stack	Vite + React 18 + TypeScript + Tailwind CSS + framer-motion
Hosting	Vercel (Hobby plan, Fluid Compute enabled)
Auto-deploy	Push ke main → Vercel build + deploy ~1 menit
PWA features	Service worker (Workbox), manifest, offline cache
Mobile-first	Iya — primary target Matthew gunakan dari iPhone/iPad

2.3 AI Department (Struktur Organisasi)



3. Arsitektur Saat Ini (Live)

3.1 Diagram Tingkat Tinggi



3.2 Komponen Per Layer

Frontend (Vite + React): Path `src/`, entry `src/main.tsx`, routes `/`, `/atmaja`, `/inbox`. Key components `Atmaja.tsx` (chat dengan library drawer), `Inbox.tsx` (briefs list). Lib helpers `atmajaClient.ts`, `atmajaFilesClient.ts`, `briefStore.ts`, `learningMemory.ts`.

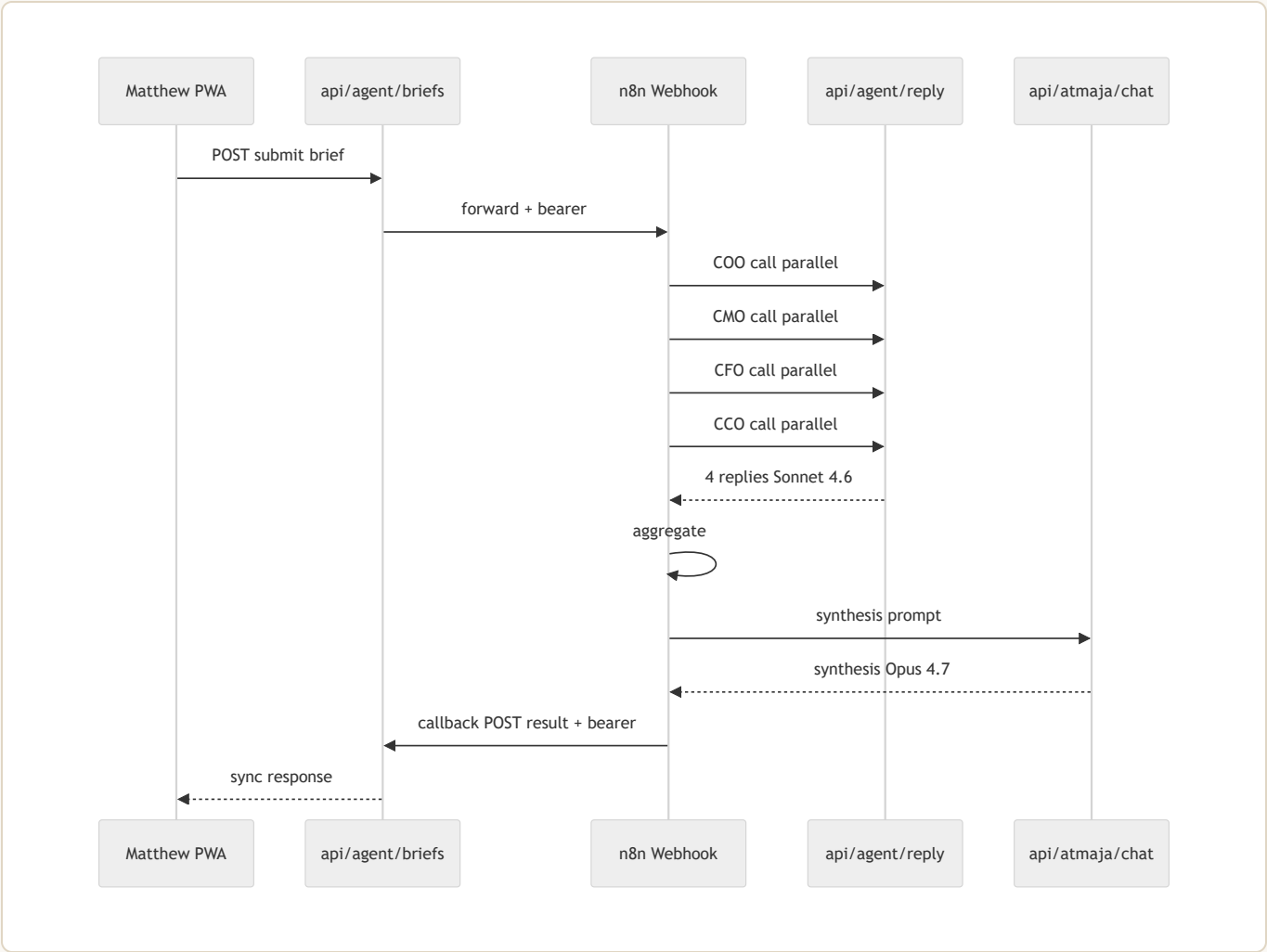
Backend (Vercel Serverless Functions): Total **12 functions** (Hobby plan cap). Daftar lengkap di section 8.

AI Runtime: Primary gateway OpenRouter, model floor **anthropic/claude-sonnet-4.6** minimum. Opus 4.7 untuk content tier (Atmaja synthesis), Sonnet 4.6 untuk orchestration (C-suite + memory extractor). Native PDF reading via OpenRouter file content block.

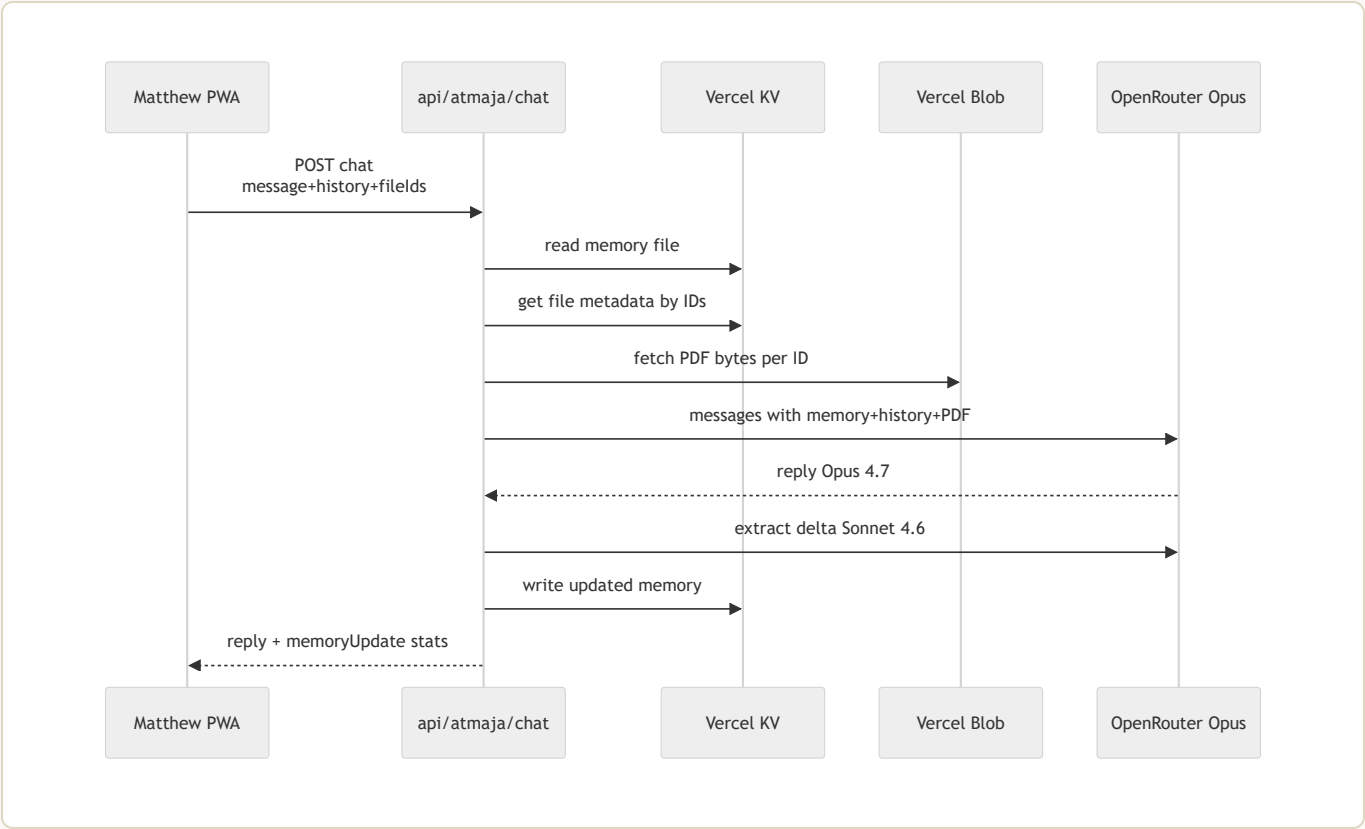
Storage: Vercel KV (Upstash Redis) untuk memory file + index + counters; Vercel Blob (private) untuk PDF binary; localStorage browser untuk chat thread + brief list per device.

Orchestration: n8n Cloud (`mahewai.app.n8n.cloud`), 2 workflow live, 3 planned. Sudah OAuth via MCP.

3.3 Data Flow: Brief Lifecycle



3.4 Data Flow: Atmaja Chat dengan Memory + PDF



4. Memory Stack 3-Lapis

Atmaja punya memory yang efektif kayak Claude.ai — terus berkembang dan tidak hilang antar sesi. Implementasi: 3 lapis terpisah dengan tanggung jawab berbeda.

4.1 Lapis 1 — Working Memory (Per Session)

Property	Value
Storage	localStorage browser + payload chat
Capacity	100 pesan terakhir × 15.000 karakter per pesan
Persist	Survives close tab/browser/OS restart (browser yang sama)
Cross-device	TIDAK (per device)
Implementation	<code>api/atmaja/chat.js</code> <code>normalizeHistory()</code> + <code>src/lib/atmajaClient.ts</code> <code>stripHistory()</code>

Effect: Dalam 1 hari kerja, Atmaja tidak lupa konteks yang baru dibahas di session aktif.

4.2 Lapis 2 — Long-term Memory File (Cross-Session)

Property	Value
Storage	Vercel KV (Upstash Redis)
Key	<code>atmaja:memory:matthew</code> (+ <code>:version</code> + <code>:turns</code>)
Format	Markdown file, sectioned by topic
Capacity	50.000 karakter total
Auto-update	Iya — tiap turn substantive, Sonnet 4.6 extract delta + append
Skip trivial	User < 30 char ATAU Atmaja reply < 200 char → skip extractor
Persistence	Permanent across device/browser/session
Endpoint	<code>GET/PUT/DELETE /api/atmaja/memory</code>

4.3 Lapis 3 — File Library (PDF Persistence)

Property	Value
Storage	Vercel Blob (private store)
Index	Vercel KV key <code>atmaja:files:matthew:index</code>
Capacity	50 files max × 2.6 MB raw PDF each
Auth fetch	SDK get() dengan access:'private' + BLOB_READ_WRITE_TOKEN
Endpoint	<code>GET/POST/DELETE /api/atmaja/memory?type=files</code>
Reference dari chat	Payload field <code>attachedFileIds: ['file_xxx']</code> (max 3 per turn)
Auto-metadata	page count (regex parser), uploaded timestamp, size, description
Memory integration	Auto-append entry ke section "## Files / Dokumen"

5. Security Architecture

5.1 Auth Pattern (api/_shared.js)

Semua 12 endpoint pakai shared helper `isRequestAllowed(req)` :

- 1. **Path 1:** Bearer token valid → allowed (server-to-server, admin curl, n8n callback)
- 2. **Path 2:** Browser same-origin (gerai.mahework.com) atau configured allowlist → allowed
- 3. **Else:** 403 reject dengan reason

5.2 Endpoint Hardening (All 12 Endpoints)

Endpoint	Method	Auth	Rate Limit
<code>/api/atmaja/chat</code>	POST	bearer OR same-origin	12/min/IP
<code>/api/atmaja/memory</code>	GET/PUT/DELETE	bearer OR same-origin	—
<code>/api/atmaja/memory?type=files</code>	POST/GET/DELETE	bearer OR same-origin	—
<code>/api/agent/reply</code>	POST	bearer OR same-origin	24/min/IP
<code>/api/agent/briefs</code>	POST	bearer OR same-origin	20/min/IP
<code>/api/agent/briefs?action=result</code>	POST	bearer only	30/min/IP
<code>/api/agent/briefs?action=list</code>	GET	bearer only	—
<code>/api/agent/health</code>	GET	public	—
<code>/api/knowledge/sync</code>	POST/GET	bearer	20/min/IP
<code>/api/openai/image</code>	POST	bearer OR same-origin	6/min/IP
<code>/api/openai/video/*</code>	various	bearer OR same-origin	varies
credits endpoints	GET	bearer OR same-origin	—

5.3 Content Security Policy

`vercel.json` configures CSP strict: `default-src 'self'` , no inline scripts, no external CDNs in connect-src, frame/object/form-action all none.

6. AI Personas

6.1 Atmaja CEO

Property	Value
Role	CEO AI Department, partner berpikir Matthew
Model	anthropic/claude-opus-4.7 (content tier)
Fallback	anthropic/claude-sonnet-4.6
Max tokens per call	8.192
Auto-continuation	Iya — sampai 24.576 tokens total (3 calls) per turn
Vision	Aktif (JPG/PNG/WEBP/GIF)
PDF native reading	Aktif via OpenRouter file content block
Long-term memory	Read tiap call, auto-update tiap turn
System prompt	Bahasa Indonesia, ringkas langsung, brand canon Gerai

6.2 4 C-suite + 12 Specialist

Semua di tier orchestration (Sonnet 4.6). Endpoint `/api/agent/reply`, route by `role` parameter. 4 C-suite: COO/CMO/CFO/CCO. 12 specialist: HR Systems, Production Manager, Curator, Brand Strategist, Market Researcher, Sales Strategist, Innovation Scout, Business Designer, Financial Analyst, Document Writer, Editorial, Web Researcher.

7. n8n Workflow Orchestration

7.1 Workflow #1 — Brief Lifecycle (LIVE v3)

Property	Value
ID	hrhkPoi0N79o07un
Trigger	POST webhook https://mahewai.app.n8n.cloud/webhook/gerai-brief-submit
Nodes	11 + 3 sticky note
Body shape	Direct atau payload-wrapped (dual support)
Retry	3× delay 5s di semua HTTP node
Smoke test result	Pass 30.3s end-to-end
Cost per execution	~\$0.03 (4× Sonnet 4.6 + 1× Opus 4.7)

7.2 Workflow #2 — Daily Digest (LIVE, channel pending)

Property	Value
ID	mUdhpkz8zmNVDbM6
Trigger	Cron <code>0 0 23 * * *</code> UTC = 07:00 WITA
Nodes	6: Schedule → Fetch Briefs → Build Prompt → Skip If Empty → Atmaja Digest → Format Output
Output channel	BELUM ADA (pending Matthew pilih)
Cost per execution	~\$0.005 (1× Opus 4.7)

7.3 Planned (Belum dibuat)

Workflow	Trigger	Purpose	Effort
#3 Notion Knowledge Sync	Notion webhook	Pull brand canon + SOP dari Notion ke server cache	1-2 jam
#4 Image Generation Batch	Manual/cron	Generate catalog images via gpt-image-2	2-3 jam
#5 Market Radar	Cron Senin 06:00 WITA	Pull kompetitor + tren via Tavily/Brightdata	2-3 jam

9. Roadmap & Arah Pengembangan

9.1 Short-term (1-2 minggu) — Quick Wins

Priority	Item	Effort	Cost impact
1	2FA Vercel + n8n + GitHub	15 menit total	Free
2	Channel output Daily Digest — pilih: callback app endpoint, atau email/Telegram/PWA push	1-2 jam (callback paling cepat)	Free-\$5/bulan
3	Rotate token N8N_WEBHOOK_TOKEN baru	30 menit	Free
4	Workflow #3 Notion Knowledge Sync	1-2 jam	Free (Notion gratis)

9.2 Medium-term (1-3 bulan) — Capability Expansion

Item	Effort
Workflow #4 Image Gen Batch — bulk product catalog images via gpt-image-2	2-3 jam
Workflow #5 Market Radar — Tavily/Brightdata pull Senin 06:00	2-3 jam
Persona hot-swap — brand canon di KV/Notion, bukan hardcoded di kode	1 hari
Cost tracking dashboard — real-time di PWA	1 hari
Backup memory file — daily auto-backup ke Blob	4 jam
Multi-thread chat per topik — strategi/vendor/brand terpisah	2 hari

9.3 Long-term (3-12 bulan) — Strategic Expansion

Item	Effort	Notes
Vector DB + RAG — Upstash Vector untuk semantic search memory + PDF	1 minggu	Free tier 10K vectors
Image generation aktif di PWA — quick command /image	3 hari	\$0.04/image
Sora 2 video gen di PWA	3 hari	\$1.00/4s video
Multi-language EN/ID untuk customer luar	1 minggu	Free
Storefront e-commerce dengan payment	2 minggu	Butuh Stripe/Xendit
Upgrade Vercel Hobby → Pro	Instant	\$20/bulan, > 12 functions

10. Gap Analysis — Yang Masih Kurang

10.1 Security Gaps

Gap	Risiko	Severity	Fix
2FA belum aktif di Vercel + n8n + GitHub	Account compromise = token leak	HIGH	Aktifkan 2FA (5 menit)
1 token forever, belum ada rotation policy	Kalau leak, exposure permanent	MEDIUM	Rotate per 3 bulan
Memory file tidak ada backup	Data loss kalau KV down	MEDIUM	Daily backup ke Blob
Tidak ada audit log siapa akses apa kapan	Tidak ke-detect incident	LOW	Vercel logs ada, enrich kalau perlu
Brand canon hardcoded di code	Update brand = git push + redeploy	LOW	Workflow #3 Notion sync

10.2 Operational Gaps

Gap	Severity	Fix
Daily Digest tidak ada channel output	MEDIUM	Pilih channel (rekomendasi callback)
Tidak ada cost dashboard real-time	MEDIUM	Build di PWA
Tidak ada error alerting	LOW	Enable Vercel notifications
Tidak ada monitoring uptime external	LOW	Better Uptime / Uptime Robot free

10.3 Feature Gaps

Gap	Severity	Fix
Cross-device sync	MEDIUM	Memory KV sudah cross-device, tinggal thread sync
Multi-thread chat per topik	LOW	UI redesign PWA
Search di memory file	LOW	Pagination atau Vector DB
Bulk PDF upload	LOW	Multi-file picker
Image generation di PWA UI	MEDIUM	Endpoint ada, UI belum

10.4 Architectural Gaps

Gap	Severity	Fix
Hobby plan = 12 functions limit (sudah mentok)	MEDIUM	Upgrade Pro \$20/bulan atau konsolidasi lebih agresif
Tidak ada test suite	MEDIUM	Vitest unit tests per endpoint
Tidak ada CI/CD checks	LOW	GitHub Actions pre-merge

11. Skill Assistant (Claude) yang Tersedia

11.1 Akses Infra Gerai

Akses	Capability
Repo Gerai (PC ini)	Read + edit + commit + push ke MahewAi/Mahew main branch
Vercel project	Trigger redeploy via push, monitor via API curl
Vercel KV	Memory storage + file index R/W
Vercel Blob	PDF storage R/W via SDK
n8n MCP	Create/update/test/publish workflow via SDK (OAuth-ed)
OpenRouter	Opus 4.7 + Sonnet 4.6 runtime
GitHub	Git operations, gh CLI untuk PR/issue

11.2 Engineering Capabilities

- TypeScript/JavaScript editing (React, Node, ES modules)
- Vite + React 18 patterns
- Serverless function design (Vercel runtime)
- API design (REST, auth, rate limiting)
- Database integration (KV, Blob)
- LLM integration (OpenRouter, Anthropic patterns)
- Security audit + hardening
- Testing (typecheck, smoke test via curl)
- Git workflow

11.3 Domain Skill Ter-load (Claude Code Skills)

Relevan untuk Gerai: engineering:* (architecture, code-review, debug, testing-strategy, incident-response, tech-debt); brand-voice:* (discover-brand, enforce-voice, generate-guidelines); design:* (design-critique, design-system, ux-copy, accessibility-review); marketing:* (brand-review, campaign-plan, draft-content, email-sequence, seo-audit); product-management:* (write-spec, roadmap-update, brainstorming); sales:* (call-prep, account-research, draft-outreach); searchfit-seo:* (SEO audit); productivity:* (task-management, memory-management).

Available kalau scope expand: customer-support:*; human-resources:*; finance:*; legal:*; operations:*; claude-api, update-config, security-review.

11.4 MCP Plugin Available

MCP	Status	Use case Gerai
n8n	OAuth ✓ (in use)	Workflow automation
Notion	Available	Workflow #3 Knowledge Sync, brand canon source of truth
ClickUp	Available	Task management Gerai expansion
Sanity	Available	CMS alternatif
Figma	Available	Design file integration
Box	Available	Document storage alternatif
Postiz	Available	Social media posting IG/Twitter
Cloudinary	Available	Image CDN + transformation
Brightdata	Available	Web scraping Workflow #5 Market Radar
Zoom	Available	Door Expert konsultasi via Zoom (per BP bab 6)
Searchfit SEO	Available	SEO audit komprehensif
Adobe	Available	Creative tools (Photoshop, Illustrator)

11.5 Yang TIDAK Saya Punya

- Akses Vercel dashboard UI (saya tidak bisa klik tombol, Matthew yang lakukan)
- Akses email Matthew
- Akses payment gateway
- Akses social media accounts (perlu connect via MCP)
- Cross-device memory (saya context per PC ini)
- Authority untuk approve/reject decisions — saya advisor, Matthew yang decide

13. Lampiran

A. Cost Estimation Bulanan

Component	Monthly cost
Vercel Hobby plan	\$0
Vercel KV free tier	\$0
Vercel Blob storage (1 GB)	\$0.15
OpenRouter Atmaja Opus 4.7	\$15-30
OpenRouter Sonnet 4.6 (extractor + C-suite)	\$5-15
OpenAI Image gen	\$0-20 kalau dipakai
OpenAI Video gen Sora 2	\$0-50 kalau dipakai
n8n Cloud Starter	\$0 kalau <5K execution/bulan
GitHub private repo	\$0
Domain mahewwork.com	\$1/bulan
Total realistic	\$25-45/bulan
Worst case heavy AI	\$80-100/bulan

B. Quick Reference Card

Bookmark ini:

- PWA: gerai.mahewwork.com/atmaja
- Vercel: vercel.com/mahewais-projects/gerai-app
- n8n: mahewai.app.n8n.cloud
- GitHub: github.com/MahewAi/Mahew

Dokumen disusun oleh Claude (Opus 4.7) sebagai bagian dari kerjaan setup AI Department Matthew.
Untuk update versi dengan info terbaru: minta Claude session aktif untuk regenerate.
Versi 1.0 · 24 Mei 2026 · Production state snapshot.

